

SHAWKY MOHAMED SHAWKY ABDELAZIZ

Data Scientist | Data Analyst | Machine Learning Engineer

Damietta, Egypt | Phone: 01017519107 | Email: shawkymmarya4@gmail.com

GitHub: github.com/shawkymmarya | LinkedIn: [linkedin.com/in/shawkymmarya-bb3352352](https://www.linkedin.com/in/shawkymmarya-bb3352352) | Portfolio: shawkymmarya.github.io

PROFESSIONAL SUMMARY

Results-driven Artificial Intelligence student specializing in Data Science, with a GPA of 3.96/4.00 and ranked 1st in class for four consecutive semesters. Strong foundation in Data Analytics, Machine Learning, and Business Intelligence, with hands-on experience in data preprocessing, exploratory data analysis, feature engineering, model development, and evaluation. Proficient in Python, Scikit-learn, XGBoost, SQL, SQL Server, Power BI, Tableau, and Excel. Experienced in building data-driven solutions and transforming raw data into meaningful insights and actionable business outcomes. Completed a 144-hour Data Analytics & BI program at Information Technology Institute (ITI), covering Data Warehousing, Advanced SQL, Statistics, Power BI, Tableau, and Generative AI. Recognized as a Top Performer in a 4-month Machine Learning diploma program and selected as a participant in NASA Space Apps.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++

Data Analysis & Preprocessing: Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Preprocessing, Feature Engineering, Statistical Analysis

Machine Learning: Scikit-learn, XGBoost, Ridge Regression, Logistic Regression, KNN, SVM, SMOTE, Cross-Validation, Hyperparameter Tuning, Model Evaluation

Machine Learning Concepts: Classification, Regression, Ensemble Methods, Feature Selection, Class Imbalance Handling Data

Visualization & Business Intelligence: Microsoft Power BI, Tableau, Matplotlib, Seaborn, Interactive Dashboards, Data Visualization, Business Intelligence

Databases & SQL: SQL, MySQL, SQL Server, Advanced SQL Queries, ER Modeling, Database Normalization (1NF, 2NF, 3NF), Data Warehousing Fundamentals

Excel & Analytics: Microsoft Excel, Data Analysis, Data Exploration

Generative AI: Generative AI, AI Agents, Workflow Automation

Tools & Environments: Jupyter Notebook, VS Code, MySQL Workbench

SOFT SKILLS

- Communication & Presentation
- Teamwork & Collaboration Analytical
- Thinking & Problem-Solving
- Critical Thinking
- Time Management
- Adaptability & Continuous Learning

PROJECTS

House Prices Prediction — Ridge Regression | Python, Scikit-learn, Pandas github.com/shawkymmarya/House-Prices-Prediction-ML

- Built a Ridge Regression model on 1,460+ property records, achieving $R^2 = 0.92$, MAE ~13,500, and RMSE ~18,900 — outperforming baseline linear regression by ~15% in prediction accuracy.
- Applied end-to-end feature engineering and L2 regularization across 79 input features, reducing overfitting and improving generalization on unseen data.

Titanic Survival Prediction — Logistic Regression, SMOTE | Python, Scikit-learn github.com/shawkymmarya/Titanic-Survival-Prediction-Logistic-Regression

- Developed a binary classification model achieving ~81% Accuracy, ~82% Precision, ~75% Recall, and ~78% F1 Score on the Titanic dataset (891 records).
- Applied SMOTE oversampling to address class imbalance, improving minority-class recall by an estimated 10–15% compared to an unbalanced baseline.

Bank Marketing Prediction — XGBoost | Python, XGBoost, Scikit-learn github.com/shawkymmarya/bank-marketing-xgboost

- Predicted customer subscription to term deposits on a 45,000+ record dataset using XGBoost, achieving ~87% Accuracy after systematic hyperparameter tuning.
- Reduced false positive rate through grid-search optimization, improving model precision by ~8% over default parameters.

Adult Income Classification — SVM | Python, Scikit-learn github.com/shawkymarya/adult-income-svm-classification

- Classified income levels (above/below \$50K) on the 48,000+ record UCI Adult dataset using SVM, achieving ROC-AUC $\approx$ 0.88–0.89 — indicating strong discriminative power.

Customer Churn Prediction — KNN | Python, Scikit-learn github.com/shawkymarya/customer-churn-prediction-knn

- Predicted customer churn using KNN with feature scaling and cross-validation, achieving 79% Accuracy, enabling targeted retention strategies that could reduce churn-related revenue loss.

Car Price Prediction — KNN Regressor | Python, Scikit-learn github.com/shawkymarya/car-price-prediction-knn

- Developed a KNN regression model to predict vehicle market prices, achieving MAE: 1,101, RMSE: 1,514, and R^2 : 0.85 — demonstrating high predictive reliability across diverse vehicle categories.

Hotel Reservation System Database — MySQL, SQL github.com/shawkymarya/hotel-management-system

- Designed and implemented a fully normalized relational database (3NF) for a Hotel Reservation System, encompassing 6+ interconnected entities (rooms, reservations, payments, guests) with optimized SQL queries.

EDUCATION

Horus University – Egypt | Expected Graduation: 2028

Bachelor of Science – Artificial Intelligence & Informatics | Major: Intelligent Data Science

- GPA: 3.96 / 4.00 | Ranked 1st in class for 4 consecutive semesters | Awarded Top Student Certificate 4 times

EXPERIENCE & ACTIVITIES

NASA Space Apps Cairo 2025 – Global Hackathon Participant | Cairo, Egypt, 2025

- Collaborated within an international team to design and develop AI-based solutions under a 48-hour time constraint, competing among 10,000+ participants across 160+ countries in NASA's annual global hackathon.

InnovEgypt Program – ITIDA & TIEC

- Completed 45 hours of structured training in innovation, entrepreneurship, and business model development; pitched an AI & Data Science startup concept, progressing through the program's full ideation-to-pitch pipeline.

CERTIFICATIONS & TRAINING

Data Analytics & BI (Power BI & Tableau) – Information Technology Institute (ITI)

- Completed a 144-hour professional training program covering Data Warehousing, Advanced SQL, Statistics, Power BI, Tableau, Excel for Data Analysis, Generative AI, AI Agents, and Workflow Automation.

Artificial Intelligence & Machine Learning Diploma – Route Academy

- Completed a 4-month (120+ hour) intensive program covering Machine Learning, Data Analysis, Data Visualization, and Deep Learning fundamentals; graduated as a Top Performer in the cohort.

Introduction to Data Science – Cisco Networking Academy

- Completed a foundational course covering Data Analysis, Data Visualization, Statistics, and Machine Learning fundamentals.